Automatic Detection of Light Verb Constructions in German Legal Texts

Final Project Report, CAS Natural Language Processing

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Models: boj-per/fvg-gbert-base, boj-per/fvg-pair-classifier-v2

Application: https://huggingface.co/spaces/boj-per/fvg-demo

Abstract

This report describes the development of an automatic detection system for light verb constructions (LVCs) in German legal texts, an NLP task of relevance to computational legal linguistics. Using a corpus of 41,654 documents from the Swiss Federal Tribunal (BGer), two complementary BERT-based models were trained on top of deepset/gbert-base: (1) an LVC Tagger using token classification with distant supervision (F1 = 0.348 on the gold test set), and (2) a Pair Classifier reformulating detection as binary classification over candidate verb-noun pairs (F1 = 0.777). The pair classifier more than doubles the tagger's F1, demonstrating that task reformulation can matter as much as model architecture. The project also shows that large language models (Claude Haiku 4.5) can serve as cost-efficient annotators for distant-supervision data, and provides a cautionary case study in the dangers of evaluating NLP models exclusively on automatically annotated data.

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1. Introduction

1.1 Light Verb Constructions in German

Light verb constructions (LVCs), known in German as *Funktionsverbgefüge* (FVG), are multi-word expressions consisting of a semantically bleached ("light") verb and a nominal component - typically a nominalized verb or an abstract noun, optionally within a prepositional phrase. While the light verb contributes grammatical information (aspect, voice, valency), the nominal component encodes the core predicate meaning. Together, the LVC functions as a single predicate.

LVCs in German, although some fixed combinations such as *Beschwerde erheben* exist, form an open and productive category. The linguistic literature contains no exhaustive list of LVCs. Moreover, from a grammatical perspective an LVC construction does not differ in any way from ordinary noun-verb combinations (i.e., objects in the grammatical sense) or preposition-noun-verb combinations. LVCs therefore cannot be identified purely on the basis of formal characteristics. The specialist literature does, however, describe linguistically relevant criteria for distinguishing LVCs from "regular" combinations, although these criteria are neither necessary nor sufficient and can at best serve as approximations:

- Semantic bleaching of the verb: The verb often carries no relevant meaning and merely serves as a "support" for the noun, which carries the actual meaning of the expression.
- Nominalization paraphrase: An LVC can often be paraphrased by a simple verb form (*eine Entscheidung treffen* → *entscheiden*).
- Restricted modifiability: The nominal component often cannot be expanded, for example by adjectives.
- Discontinuity: The verb and nominal component can be separated by numerous intervening elements.

This makes the automatic analysis of LVCs challenging, since such analysis cannot operate, or can operate only to a limited extent, on the formal characteristics of expressions.

Table 1 shows representative German LVC examples.

LVC	Meaning	Light Verb	Nominal Component
Beschwerde erheben	to file a complaint	erheben	Beschwerde
in Erwägung ziehen	to consider	ziehen	in Erwägung
zur Anwendung kommen	to be applied	kommen	zur Anwendung
Gehör gewähren	to grant a hearing	gewähren	Gehör
Urteil fällen	to render a judgment	fällen	Urteil
Antrag stellen	to submit a request	stellen	Antrag

Table 1: Representative German light verb constructions with English equivalents.

1.2 Light Verb Constructions in Legal Language

Legal German exhibits a particularly high density of LVCs, reflecting structural features of legal writing:

- Nominalization preference: legal events are conventionally encoded as nouns (*Klageerhebung*, *Beschwerdeführung*), making LVCs the natural predication form.
- Aspectual distinctions: LVCs encode fine-grained aspect (*in Kraft treten* [ingressive] vs. *in Kraft sein* [stative]).
- Register formality: LVCs are markers of formal, elevated register.
- Conventionalization: certain LVCs are near-obligatory in judicial language (*Beschwerde erheben, Antrag stellen, Stellung nehmen*).

Practical motivations for automatic LVC detection include: building domain-specific lexicons, corpus-based study of how legal language patterns vary across courts or time periods, and supporting downstream NLP tasks such as machine translation (where LVCs often require special handling).

1.3 Project Goals

This project addresses the following practical engineering qüstions:

- Can a BERT-based model detect LVCs in German legal text using only a reference list and unlabeled corpus data (distant supervision), without manually annotated training data?
- Which model architecture - token-level classification or pair-level binary classification - is better suited to this task?
- Can a large language model (Claude Haiku 4.5) serve as a cost-efficient annotator for distant-supervision data?

Broader linguistic questions - such as how LVC density varies across legal domains, or what distinguishes LVCs from other multi-word expressions - remain for future work. This project focuses on engineering a reliable detection pipeline.

2. Data

2.1 Corpus

The corpus consists of two sources of Swiss legal text (Table 2). For training, a representative sample of 5,000 BGer files plus all 724 cantonal files was used. After HTML extraction, normalization, and sentence segmentation with spaCy (de_core_news_lg), the training subset yielded 829,059 sentences with 5-200 tokens.

Source	Files	Format	Description
Swiss Federal Tribunal decisions (BGer)	40’930	HTML	Multiple decades of BGer decisions
Cantonal court decisions	724	TXT	Decisions from various cantonal courts

Source	Files	Format	Description
Training subset	**5'724**	-	5'000 BGer + 724 cantonal = 829'059 sentences

Table 2: Corpus composition.

2.2 FVG Reference Lists

Three manually compiled reference resources were used for distant supervision annotation:

- liste_fvg.txt - 209 curated LVC patterns: simple noun phrases (*Beschwerde erheben, Antrag stellen*), prepositional phrases (*in Erwägung ziehen, zur Anwendung kommen*), and complex constructions (*in Wiederwägung ziehen, unter Beweis stellen*).
- verb_noun_fvg.csv - a frequency list of 300 verb-noun pairs from the corpus, used as basis for manual evaluation.
- praep_noun_verb.csv - a frequency list of 200 entries (163 after deduplication) with prepositional LVC patterns (*in Frage kommen, in Betracht ziehen*), also used as basis for manual evaluation.

2.3 Gold Annotation Dataset

A total of 1,149 sentences from the corpus were manually annotated by the project author via a custom annotation interface (offline). Each sentence was judged with respect to a highlighted verb-noun pair: LVC (1), not LVC (0), or ambiguous (?). Table 3 shows the class distribution.

Class	Sentences	Proportion
LVC (manually_FVG = 1)	386	33.6%
Not LVC (manually_FVG = 0)	759	66.1%
Ambiguous (manually_FVG = ?)	4	0.3%
Total	**1'149**	**100%**

Table 3: Gold annotation dataset class distribution.

3. Methodology

3.1 Overall Architecture

The project implements an extended detection pipeline with a manual evaluation and feedback loop. Figure 1 shows the overall architecture.

```
Corpus (HTML/TXT)
  |
  v  Step 1: Text Extraction (HTML parsing, normalization)
Raw texts (JSONL)
```

```
   |
   v  Step 2: Two-Pass Distant Supervision Annotation
 Annotated training data (BIO labels, ~116k sentences)
   |
   v  Step 3: Model Fine-Tuning (deepset/gbert-base, token classification)
 FVG Token Classifier v1
   |
   v  Step 4: Push model to Hugging Face Hub
   |
   v  Step 5 / 5b: Extract evaluation sentence candidates from corpus
   |
   v  Step 6: Manual Annotation (1,149 sentences via Gradio GUI)  ->  Gold Dataset
   |
   v  Step 7: Evaluate FVG Tagger v1 against gold labels
 Error analysis  ->  extend FUNKTIONSVERBEN list
   |
   v  Step 8: Convert gold labels to BIO format + 80/20 split
   |
   v  Step 2 (re-run) + Step 3 (re-run, extended list + gold 10x oversampled)
 FVG Token Classifier v3  +  Step 9: Push to HF Spaces
   |
   v  Step 10: Test syntactic post-filter (spaCy)
   |
   v  Step 11: Prepare pair data from gold annotation
   |
   v  Step 12: Train Pair Classifier v1 (gold pairs only, F1=0.759)
   |
   v  Step 13: Evaluate Claude Haiku 4.5 as annotator
   |
   v  Step 14: Claude Haiku 4.5 annotates 37,841 DS pairs  (~$45)
   |
   v  Step 15: Train Pair Classifier v2 (gold + Claude-annotated DS, F1=0.777)
   |
   v  Gradio Demo Application (Hugging Face Spaces)
 Interactive comparison: FVG Tagger  vs.  Pair Classifier v2
```

Figure 1: Project pipeline with manual evaluation feedback loop and iterative improvement.

3.2 Annotation Scheme

Token classification uses a BIO (Beginning-Inside-Outside) scheme adapted for LVCs, with four labels (Table 4). The scheme supports discontinuous LVCs where light verb and nominal component are separated by intervening tokens.

Label	Meaning	Example
B-VERB	Beginning of the light verb	*erhebt in "Beschwerde erhebt"*
B-NOM	Beginning of the nominal component	*Beschwerde; or in/zum (for prepositional LVCs)*
I-NOM	Continuation of the nominal component	*Erwägung (after in), Schluss (after zum)*
O	Outside any LVC	all other tokens

Table 4: BIO annotation labels. Discontinuity example: "Er nimmt die Einwände in Betracht ." receives labels O B-VERB O O B-NOM I-NOM O

3.3 Two-Pass Distant Supervision Annotation

Distant supervision avoids the need for manually labeled training data by using the FVG reference list to automatically annotate the corpus. Two complementary passes are applied:

Pass 1 - List-Based (High Precision)

For each of the 209 LVC patterns, the corpus is searched for co-occurrences of the light verb lemma and nominal component lemma(s) within a window of +/-20 tokens. Lemmatization with spaCy enables detection of all inflected forms (*erhebt, erhob, erhoben* → lemma *erheben*). Matched sentences receive BIO annotations.

Pass 2 - Syntax-Based (Generalization)

Using the spaCy dependency parser, sentences are analyzed for verb-noun pairs where the verb belongs to the FUNKTIONSVERBEN set (~35 verbs):

- Direct object (dep = oa): a light verb governing an abstract noun as direct object is annotated as LVC.
- Prepositional phrase (dep = mo, cvc): a light verb governing a PP with an LVC-typical preposition (*in, zu, auf, an,* etc.) is annotated as LVC.

Verbs *haben* and *sein* are explicitly excluded from FUNKTIONSVERBEN to avoid false annotation of subjects as nominal components. Following manual evaluation (Section 3.5), the list was extended from 28 to 35 verbs:

Verb added	Reason
vornehmen	41 false negatives - most frequent error type
erteilen	35 false negatives
aussprechen	10 false negatives
belegen, schliessen	6 false negatives each
anordnen, einreichen	Frequent legal LVC verbs identified in error analysis

Table 5: Verbs added to FUNKTIONSVERBEN set after FVG Tagger v1 error analysis.

Negative examples are sampled at a 2:1 ratio (negative:positive) to balance training data.

3.4 Model Fine-Tuning

3.4.1 FVG Tagger (Token Classification)

The FVG Tagger is a token classification model (deepset/gbert-base) fine-tuned to assign BIO labels to each token. Training parameters are listed in Table 6.

Parameter	Value
Base model	deepset/gbert-base
Task	Token Classification (4 labels: B-VERB, B-NOM, I-NOM, O)
Epochs	4

Parameter	Value
Batch size	32
Learning rate	2e-5
Weight decay	0.01
Warmup	10% of steps
FP16	Yes (CUDA)
Best-model criterion	F1 score on development set
Gold oversampling	10x (299 gold sentences, each repeated 10 times)
Subword alignment	B-NOM/I-NOM subwords → I-NOM; B-VERB subwords → -100 (ignored in loss)

Table 6: FVG Tagger training parameters.

3.4.2 Pair Classifier (Sequence Classification)

After the FVG Tagger showed limited F1 on the gold test set (Section 4.2), the second project phase reformulated detection as binary sequence classification. For each candidate verb-noun pair extracted from a sentence, the classifier predicts LVC / not-LVC.

> Input format: [CLS] sentence [SEP] verb pattern [SEP]
>
> Example: [CLS] *Der Beschwerdeführer hat Beschwerde erhoben.* [SEP] *erhoben Beschwerde* [SEP]

For prepositional LVCs, the preposition is prepended to the pattern: *gezogen in Erwägung.*

Two versions were trained: v1 on gold pairs only (932 train / 244 test), and v2 on gold pairs plus Claude Haiku 4.5-annotated DS pairs with differential sample weights to account for data quality (Section 4.3).

3.5 Manual Evaluation and Gold Labels

Evaluation against automatically annotated data gives inflated performance metrics. To measure true performance, a gold standard was created through manual annotation.

Sentence Extraction

Verb-noun pairs were extracted from the three reference lists, and for each pair, up to 3 corpus sentences containing both elements were retrieved: 810 for simple verb+noun pairs (from verb_noun_fvg.csv) and 339 for prepositional LVC patterns (from praep_noun_verb.csv), totaling 1,149 sentences.

Annotation Protocol

The annotation interface (custom GUI, offline) presented each sentence with the candidate pair highlighted. The annotator judged whether the pair forms an LVC in the given sentential context (1/0/?). The judgment was context-dependent: a verb and noun that co-occur but are not syntactically related - or where a different expression forms the actual LVC - were labeled 0.

Gold Label Integration into Training

- Training set (80% of LVC=1 sentences): 299 sentences converted to BIO format and added with 10x oversampling to give them stronger influence despite small count.
- Held-out test set (20% of LVC=1 + all LVC=0): ~230 sentences reserved for evaluation. After pair extraction and gold test set refinement (Section 4.4), this yields 81 positive and 163 negative pairs (n=244).

Split	Positive	Negative	Total
Train (80% LVC=1 + all LVC=0)	325	607	932
Test / held-out (20% LVC=1)	92	152	244
Cleaned test set (after Section 4.4)	**81**	**163**	**244**

Table 7: Gold dataset splits for the Pair Classifier.

3.6 Syntactic Post-Filter

A rule-based syntactic post-filter (fvg_syntax_filter.py) validates each predicted pair using the spaCy dependency parse. Two checks are applied:

- Dependency relation check: the nominal component must stand in a valid LVC relation to the light verb (Table 8).
- Clause boundary check: the dependency path from verb to nominal must not cross a clause boundary - relative clause, object clause, adverbial clause (CLAUSE_BOUNDARY_DEPS = {rc, oc, neb, om, uc}). If the path crosses such a boundary, the pair is rejected.

Relation Type	Condition
Direct object	NOM.dep in {oa, obj, dobj} AND NOM.head == VERB
Prepositional object	NOM.dep == nk AND NOM.head.dep in {mo, cvc, op, pg, obl} AND NOM.head.head == VERB

Table 8: Dependency relations accepted by the syntactic post-filter. Prepositional LVCs also require the preposition to be in the whitelist: in, zu, auf, an, ausser, unter, vor, ohne, aus.

4. Results

4.1 Training Data (Final Version, Model v3)

Category	Sentences
Positive sentences (DS, with LVC)	37’841
Negative sentences (DS, without LVC)	75’993

Category	Sentences
Gold sentences (299 x 10x oversampled)	2'990
Total training	**116'824**
Development set	14'229
Test set (distant labels, for reference)	14'230

Table 9: Training data composition for FVG Tagger v3.

4.2 FVG Tagger Evaluation

The FVG Tagger achieves F1 = 0.954 on the development set (automatically annotated). When evaluated against the manually annotated gold test set, performance drops to F1 = 0.348 - nearly three times lower. This discrepancy illustrates a critical pitfall of distant supervision; the model has learned the same patterns as the annotation procedure. Its high dev-set score reflects pattern-matching skill, not genuine linguistic understanding.

Table 10 shows the agreement between distant-supervision annotations and the human gold labels:

Metric	Value
Precision (DS annotations vs. gold)	0.40
Recall (DS annotations vs. gold)	0.44
F1 (DS annotations vs. gold)	**0.42**

Table 10: Agreement between distant-supervision annotations and manual gold labels. The low F1 (0.42) reflects substantial inherent noise in the distant supervision approach.

Table 11 breaks down FVG Tagger v3 performance on the gold test set by LVC type:

Evaluation subset	n	Precision	Recall	F1
All pairs (with syntactic filter)	822	0.249	0.582	**0.348**
Simple verb + noun pairs	629	0.163	0.581	0.255
Prepositional LVCs	193	0.656	0.583	0.618

Table 11: FVG Tagger v3 gold evaluation by LVC type. Prepositional LVCs achieve higher precision (0.656) due to the preposition whitelist acting as an additional hard constraint.

Table 12 shows the impact of the syntactic post-filter:

Metric	Without filter	With filter	Change	
Precision	0.218	0.249	+14%	
Recall	0.595	0.582	-2%	
F1	0.319	**0.348**	+9%	
False Positives	169	139	-30 (-18%)	

Table 12: Impact of the syntactic post-filter on FVG Tagger v3 gold evaluation.

Error analysis: most frequent false negatives - *vornehmen* (6 FN), *erteilen* (5 FN), *fallen* (3 FN, e.g. *ausser Betracht fallen*), *lassen* (3 FN, ambiguous between full verb and light verb). Most frequent false positives (with filter) - *geben + Möglichkeit* (3 FP), *leisten + Sicherheit* (3 FP, borderline LVC status), *führen + Lebensgemeinschaft* (3 FP, full-verb reading).

4.3 Pair Classifier Results

The second project phase reformulated LVC detection as binary classification over candidate verb-noun pairs. The model receives a sentence and a specific pair as input and predicts LVC/not-LVC. Table 13 shows results on the cleaned gold test set (n=244):

Model	Precision	Recall	F1
Pair Classifier v1 (gold pairs only)	0.710	0.815	0.759
Pair Classifier v2 (gold + Claude-annotated DS)	**0.803**	**0.753**	**0.777**

Table 13: Pair Classifier results on the cleaned gold test set (81 positive / 163 negative). Both versions substantially outperform the FVG Tagger (F1 = 0.348).

The pair classifier more than doubles the FVG Tagger's F1 (0.348 -> 0.777). The performance jump is attributable to the task reformulation: instead of labeling every token in a sentence, the model evaluates a specific verb-noun pair with the pair made explicit in the input - a substantially more focused and tractable classification problem.

Adding Claude-annotated DS pairs (v2) improves precision (0.710 -> 0.803) while recall decreases slightly. This reflects Claude's conservative annotation strategy: additional training examples reinforce rejecting borderline cases, which aligns with the refined gold criteria.

Claude Haiku 4.5 was evaluated on the gold test set (n=244): Precision = 0.900, Recall = 0.500, F1 = 0.643. Its high precision and lower recall confirm a conservative approach. Notably, approximately 14 of the 45 apparent false negatives were later found to be gold annotation errors (Section 4.4), meaning Claude's effective recall is somewhat higher.

Table 14 shows the distribution of Claude's annotations on the 37,841 DS pairs:

Label	Count	Proportion
LVC (ja)	9'711	25.7%
Not LVC (nein)	28'041	74.1%
Ambiguous / unclear	89	0.2%
Total	**37'841**	**100%**

Table 14: Claude Haiku 4.5 annotations on the 37,841 DS pairs.

Pair Classifier v2 uses differential sample weights to account for data source quality:

Data source	Label	Sample weight
Gold (manual annotation)	Positive	1.0
Gold (manual annotation)	Negative	1.0

Data source	Label	Sample weight
DS (Claude-annotated)	Positive	0.5
DS (Claude-annotated)	Negative	0.3

Table 15: Sample weights in Pair Classifier v2. DS negatives are capped at 2x DS positives (19,422 sentences). Total: 30,065 training pairs (10,036 positive / 20,029 negative).

4.4 Gold Test Set Refinement

Comparing Claude's annotations with the original gold labels revealed 19 likely errors. These were reviewed by the project author and confirmed as annotation mistakes:

- 15 cases changed LVC=1 → LVC=0 (false positives in original annotation): expressions that fail the Simplex-Test because they are frozen legal phrases with no simplex paraphrase. Examples: *auf Abweisung schliessen* (x8, "to move for dismissal" - a fixed procedural formula, not a productive LVC), *von Amtes wegen* (x2), *Bedingung stellen* (x2), *Eingabe einreichen*, auf *Antrag verfolgen*.
- 4 cases changed LVC=0 → LVC=1 (missed LVCs in original annotation): *zu Grunde legen* (a duplicate instance not previously recognized), *Würdigung vornehmen*, *Konzession erteilen*, *Auslegung geben*.

This correction demonstrates that annotation is an iterative process. Even manually created gold labels contain errors, especially at borderline cases. Key insight: some expressions initially labeled as LVCs are lexicalized multi-word expressions (*auf Abweisung schliessen* = "to move for dismissal", a frozen legal formula) rather than productive LVCs. Future annotation efforts should incorporate review cycles and inter-annotator agreement measures.

4.5 Full Model Comparison

Table 16 compares all model versions. The FVG-Tagger (DS) row reflects evaluation on the automatically annotated DS pair test set; the other rows use the cleaned gold test set. Due to different evaluation setups, the DS-evaluated figure is not directly comparable to the gold-evaluated figures and is shown for reference only.

Model	Precision	Recall	F1	Eval. set
FVG-Tagger (DS labels) (dagger)	0.566	0.531	0.548	DS pair test
FVG-Tagger + spaCy filter	-	-	**0.348**	Gold held-out
Pair Classifier v1	0.710	0.815	0.759	Cleaned gold
Pair Classifier v2	**0.803**	**0.753**	**0.777**	Cleaned gold

Table 16: Full model comparison. (dagger) FVG-Tagger (DS) was evaluated on the automatically annotated DS pair test set - not the gold test set. This figure is not directly comparable to the gold-evaluated metrics in the other rows.

4.8 Demo Application

The final application (app/demo.py) compares both model approaches interactively on HF Spaces.

- Model 1 - FVG Tagger: token classifier on the full sentence. Strength: detects konjunktiv forms (e.g. *böten,* konjunktiv II of *bieten*) and coordinated structures that confuse the dependency parser. Weakness: lower precision.
- Model 2 - spaCy + Pair Classifier v2: spaCy extracts syntactic candidate pairs (direct objects, PP-objects with LVC prepositions); the pair classifier decides for each. Strength: higher precision. Weakness: fails when spaCy mis-parses the sentence structure.

The two approaches are genuinely complementary and an ensemble could improve overall performance.

5. Large-Scale Corpus Application

To validate the pipeline beyond the held-out test set and to characterize LVC usage in Swiss Federal Administrative Court decisions, the trained models were applied to the full corpus of 40,930 BGer documents.

The pipeline processed 5,159,730 sentences across all 40,930 documents. The FVG Tagger identified at least one LVC in 280,187 sentences (5.4%); the spaCy + Pair Classifier v2 pipeline found at least one LVC in 186,871 sentences (3.6%). The lower rate from the v2 pipeline reflects its higher precision: it rejects candidates that the tagger accepts, including many false positives from simple verb+noun constructions.

The lists below show the top-20 LVCs detected by each model across the corpus. Both models agree on the dominant patterns: *in Erwägung ziehen* (to consider), *Beschwerde erheben* (to file a complaint), and *Stellung nehmen* (to take a position) are among the most frequent in Swiss administrative law, consistent with domain expectations.

Pair Classifier

1. in Erwägung ziehen (8’391)
2. Beschwerde erheben (4’599)
3. Beschwerde weisen (4’029)
4. in Frage stellen (3’536)
5. Antrag stellen (3’169)
6. in Frage kommen (3’005)
7. Stellung nehmen (2’711)
8. Auffassung vertreten (2’669)
9. in Kraft treten (2’558)
10. in Kauf nehmen (2’220)

11. Rechnung tragen (2’218)
12. zum Ausdruck bringen (2’212)
13. in Betracht fallen (2’005)
14. in Betracht kommen (1’987)
15. in Frage stehen (1’889)
16. in Anspruch nehmen (1’819)
17. Urteil aufheben (1’716)
18. ausser Betracht fallen (1’662)
19. Beschwerde abweisen (1’655)
20. in Betracht ziehen (1’439)

FVG Tagger

1. Frage stellen (8’817)
2. in Erwägung ziehen (8’371)
3. Beschwerde erheben (5’899)
4. Beschwerde führen (4’686)
5. in Frage kommen (3’259)
6. Anwendung finden (3’187)
7. Gebrauch machen (2’991)
8. Stellung nehmen (2’658)
9. Antrag stellen (2’657)
10. zur Verfügung stehen (2’424)
11. zum Ausdruck bringen (2’329)
12. in Frage stehen (2’193)
13. in Kauf nehmen (2’118)
14. in Betracht kommen (2’095)
15. zum Schluss kommen (1’967)
16. in Kraft treten (1’846)
17. zur Verfügung stellen (1’836)
18. in Anspruch nehmen (1’835)
19. Gesuch stellen (1’734)
20. Klage erheben (1’572)

Several patterns in the top list illustrate both the strengths and limitations of the approach. The Pair Classifier correctly recovers domain-specific prepositional LVCs such as *fallen ausser Betracht* and *stellen in Abrede* that are not easily detected by simple form matching. However, entries such as *Beschwerde abweisen* and *Urteil aufheben* are arguably compositional: *die Beschwerde abweisen* (to dismiss the complaint) functions as a full verb construction, not a light verb one. Their high frequency in the list is a reminder that the precision of the corpus-level results is bounded by the F1 = 0.777 measured on the gold test set, and that manual spot-checking of the top patterns is advisable before drawing linguistic conclusions.

The corpus analysis confirms that LVCs are pervasive in Swiss administrative law, appearing in roughly one in every 20–28 sentences depending on the detection model, and that a small set of high-frequency patterns accounts for a disproportionate share of all occurrences; a Zipfian distribution consistent with findings in general-language LVC research.

6. Discussion

6.1 Strengths

The most significant advantage of the approach is that no manual annotation is required for training: distant supervision generates 116,824 training sentences from just 209 LVC patterns. The use of spaCy lemmatization makes the system robust to inflection, capturing all inflected forms including rare subjunctive forms that form-based matching would miss. Discontinuous LVCs, where verb and nominal component are separated by many tokens, are handled naturally by the BERT architecture combined with window-based matching. Coverage extends beyond the 209 curated reference patterns through the syntax-based Pass 2, which labels LVCs not present in the reference list. The system also benefits from iterative improvement through manual feedback: error analysis after v1 identified the most critical coverage gaps (*vornehmen, erteilen*) and addressed them by extending FUNKTIONSVERBEN and integrating gold labels. Finally, the FVG Tagger and Pair Classifier have complementary error profiles - the tagger handles subjunctive and coordination while the pair classifier achieves higher precision on standard cases - meaning an ensemble could leverage both strengths.

6.2 Limitations

The most fundamental concern is evaluation circularity: the F1 of 0.954 on the development set measures consistency with the automatic annotation rather than linguistic correctness. External evaluation reveals F1 = 0.348, which is a nearly threefold difference that illustrates how DS noise does not surface in internal evaluations. A related weakness is low precision for simple verb+noun pairs (F1 = 0.255): the model cannot reliably distinguish true LVCs from compositional constructions, such as *eine Lebensgemeinschaft führen* (full verb) versus *Verhandlungen führen* (LVC). The syntactic post-filter reduces false positives by 18% but cannot resolve this fundamental semantic ambiguity. The spaCy parser itself introduces further errors, misanalyzing complex legal sentence structures with multiple embeddings and coordination, which causes genuine LVCs to be rejected. The system is also constrained by list dependence: it is limited to verbs in FUNKTIONSVERBEN, leaving productive LVC verbs such as *fallen* (*ausser Betracht fallen*), *lassen*, and *bleiben* uncovered. On the evaluation side, the gold test set of 75 positive sentences is small enough that the metrics carry high statistical variance; confidence intervals would be methodologically appropriate. The gold labels were also produced by a single annotator, so without inter-annotator agreement, the reliability of the standard cannot be quantified.

6.3 Possible Extensions

- Extend FUNKTIONSVERBEN to cover *fallen, bleiben*, and *lassen* with appropriate noun pools.
- Add a semantic post-filter verifying the Simplex-Test programmatically as an additional validation step.
- Use a larger base model (deepset/gbert-large) for better contextual disambiguation.
- Train on the full corpus (40,930 BGer files) rather than the 5,000-file sample.
- Expand the gold dataset to 500-1,000 sentences for statistically reliable evaluation.
- Establish inter-annotator agreement on a subset with a second annotator.
- Extend to French and Italian Swiss legal texts (the Swiss Federal Tribunal issues decisions in all three official languages).
- Investigate ensemble methods combining FVG Tagger and Pair Classifier predictions.

7. Conclusion

This project demonstrates that automatic detection of light verb constructions in German legal text is feasible with distant supervision and BERT fine-tuning - and that model architecture and task formulation are decisive.

The FVG Tagger (token classification) achieves F1 = 0.348 on a held-out gold test set. Its internally measured F1 = 0.954 on automatically annotated data is substantially inflated, illustrating a critical methodological warning: distant supervision generates large training datasets, but also substantial noise that does not appear in internal evaluation results. Researchers who evaluate only against automatically annotated data may significantly overestimate real-world performance.

The decisive advance came from reformulating the task: instead of labeling all LVC tokens in a sentence, the Pair Classifier decides for each (sentence, verb, noun) triple whether an LVC is present. This reformulation more than doubled the F1 (0.348 -> 0.777). Augmenting with 29,133 Claude-annotated DS pairs improved the pair classifier further (v1: F1 = 0.759 -> v2: F1 = 0.777), as Claude's conservative annotation strategy aligns well with the refined linguistic criteria (Simplex-Test, Selektionstest).

A further methodological finding: comparing Claude annotations with gold labels revealed 19 annotation errors. Annotation is an iterative process - even manually created labels contain mistakes, particularly at borderline cases such as *auf Abweisung schliessen* (a frozen legal formula, not a productive LVC). Future projects should incorporate review cycles and inter-annotator agreement measures.

The two complementary architectures in the demo application illustrate a general finding: no single model universally solves LVC detection. For production use, Pair Classifier v2 is recommended as the primary model, with the FVG Tagger as a candidate generator in syntactically complex contexts.